Marie Curie: A Short Overview

Early life and education

Marie Curie was born Maria Sklodowska in Warsaw in 1867. In 1891 she moved to Paris and enrolled at the Sorbonne, which is part of the University of Paris. She studied physics and mathematics there and later became a professor at the Sorbonne.

In 1895 Marie Sklodowska married Pierre Curie, a French physicist who taught at the School of Physics and Chemistry in Paris. Pierre Curie became her closest scientific collaborator. Their daughter Irene Joliot-Curie was born in 1897.

Discovery of polonium and radium

Henri Becquerel discovered radioactivity in 1896 while studying uranium salts. Marie Curie chose to investigate this phenomenon for her doctoral research. She and Pierre Curie discovered the element polonium in 1898 and named it after Poland. In the same year they discovered radium, working together with the chemist Gustave Bemont. Marie Curie coined the term radioactivity to describe the emission of rays by these elements.

The 1903 Nobel Prize in Physics was awarded jointly to Henri Becquerel, Pierre Curie and Marie Curie. Marie Curie became the first woman to receive a Nobel Prize.

Career after 1906

Pierre Curie died in a street accident in Paris in 1906. The University of Paris then offered Marie Curie his professorship, and she became the first woman to teach at the Sorbonne. In 1911 she received the Nobel Prize in Chemistry for the discovery of polonium and radium. She is the only person to have won Nobel Prizes in two different sciences in this period of history.

Marie Curie founded the Radium Institute in Paris, which later became the Curie Institute. The institute was funded by the University of Paris and the Pasteur Institute. It remains a leading centre for cancer research and radiotherapy.

World War I

During World War I, Marie Curie developed mobile X-ray units to help surgeons treat wounded soldiers near the front. These vehicles became known as the petites Curies. Her daughter Irene Joliot-Curie assisted her in operating the units and trained technicians to use the equipment.

Legacy

Irene Joliot-Curie continued her mother's work at the Radium Institute. Together with her husband Frederic Joliot-Curie she won the 1935 Nobel Prize in Chemistry for the synthesis of new radioactive elements. The element curium, discovered by Glenn Seaborg, was named in honour of Marie Curie and Pierre Curie. Marie Curie died in 1934.